Computer Science

CSC380: Principles of Data Science

Nonlinear Models 1

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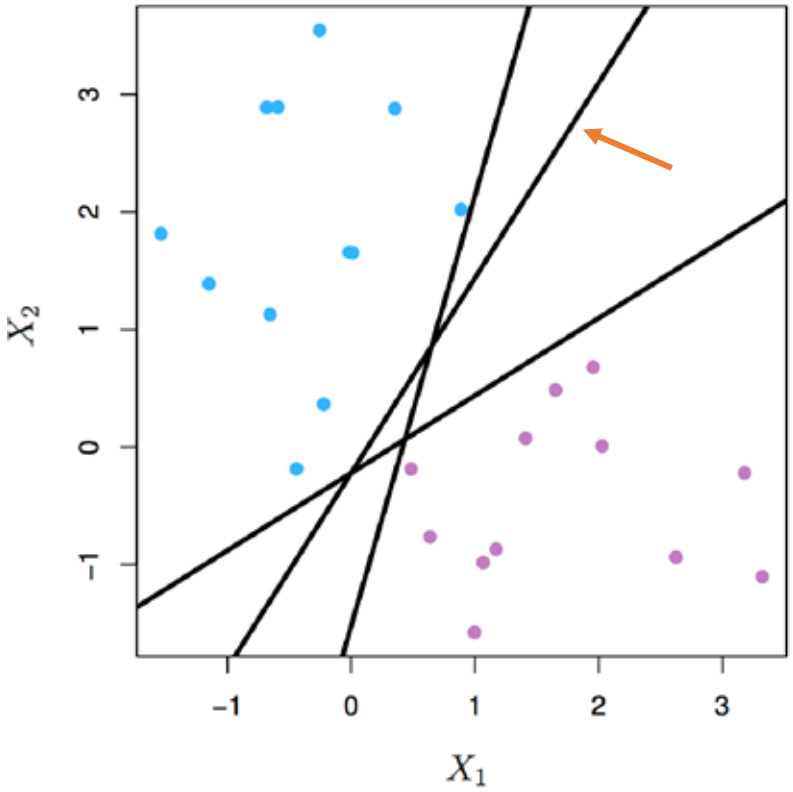
Outline

- Support Vector Machine

Linear Decision Boundary

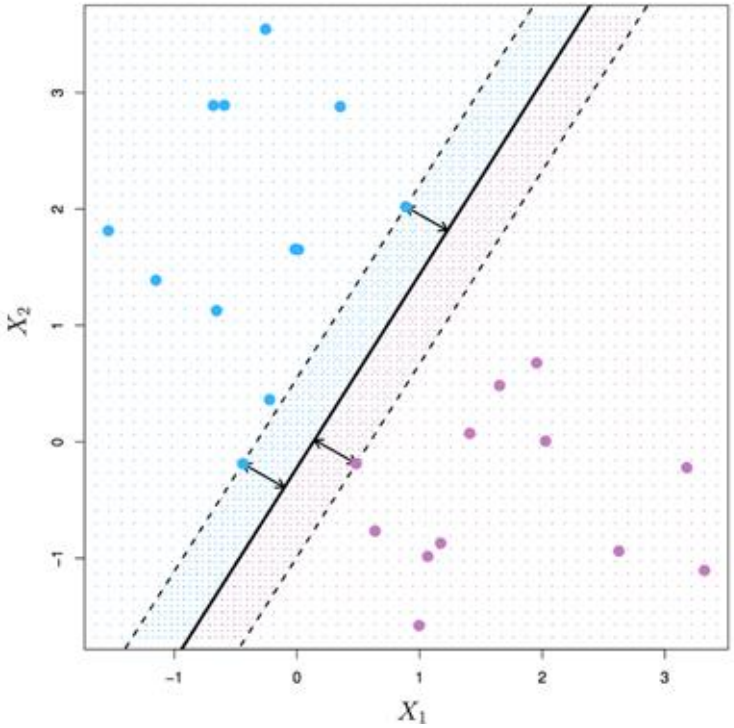
Forget about the 'regression' point of view for now..

At the end of the day, we just want a line that separates the two classes well.



Q: but if you have to choose one, which one will you choose?

Classifier Margin



*The **margin** measures minimum distance between each class and the decision boundary*

Observation Decision boundaries with larger margins are more likely to generalize to unseen data

Idea Learn the classifier with the largest margin that still separates the data…

…we call this a ***<u>max-margin classifier</u>***

Hyperplane

A linear discriminant function in D dimensions is given by a hyperplane, defined as follows:

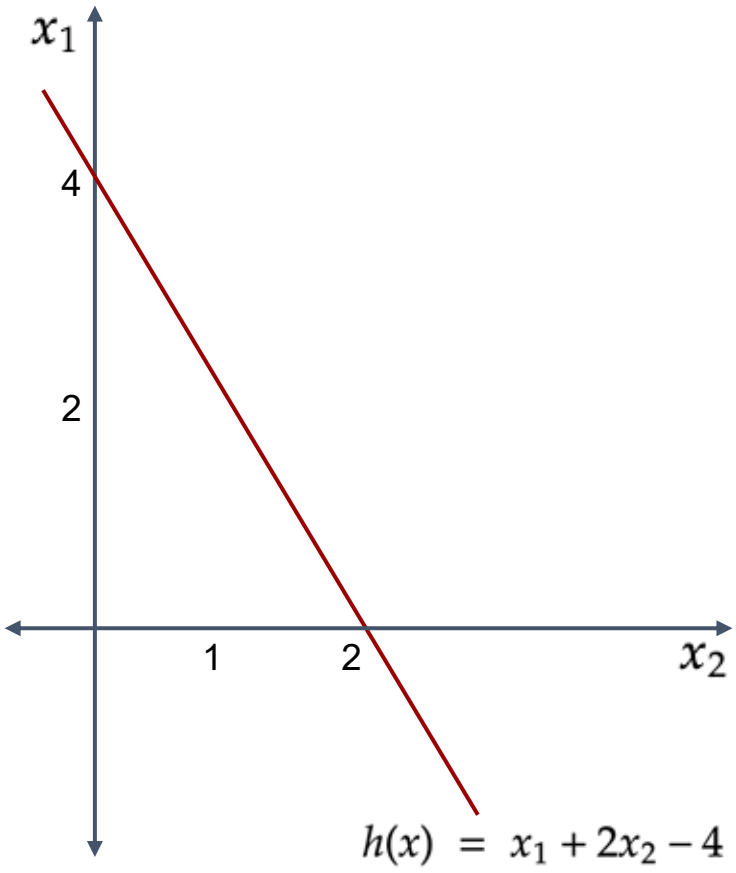
$$h(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + b$$
$$= w_1x_1 + w_2x_2 + \cdots + w_dx_d + b$$

For points that lie on the hyperplane, we have:

$$h(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + b = 0$$

Example:

$$h(x) = x_1 + 2x_2 - 4$$



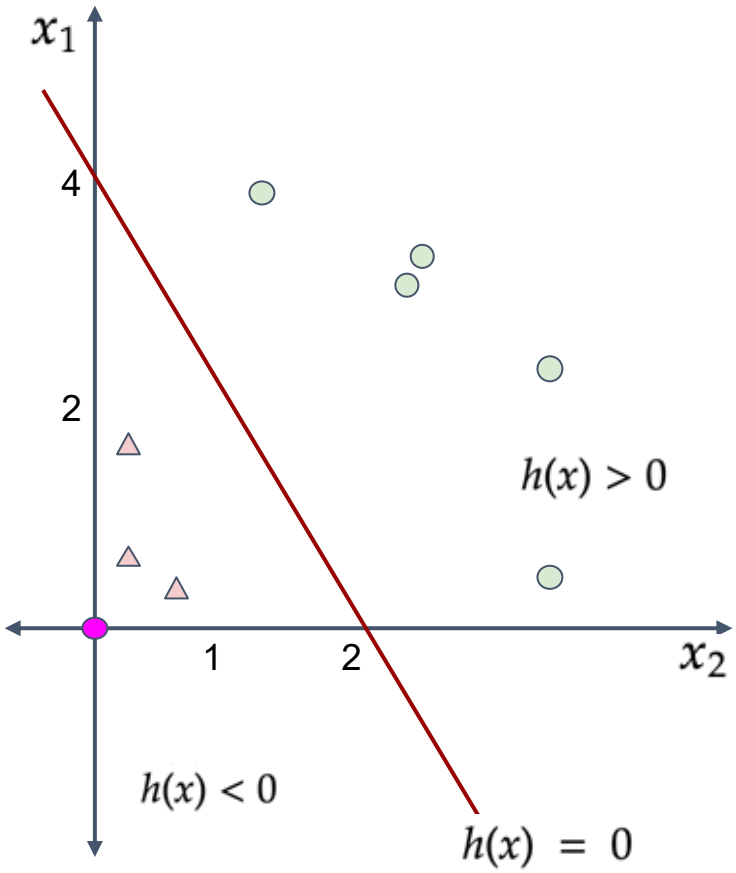
Separating Hyperplane

A hyperplane h(x) splits the original d-dimensional space into two half-spaces. If the input dataset is linearly separable:

$$y = \begin{cases} +1 & \text{if } h(\mathbf{x}) > 0 \\ -1 & \text{if } h(\mathbf{x}) < 0 \end{cases}$$

Q: what is the label for the origin?

$$h(x) = x_1 + 2 \cdot x_2 - 4 = 0 + 2 \cdot 0 - 4 = -4$$
$$y = -1$$



Separating Hyperplane: weight vector

Let a_1 and a_2 be two arbitrary points that lie on the hyperplane, we have:

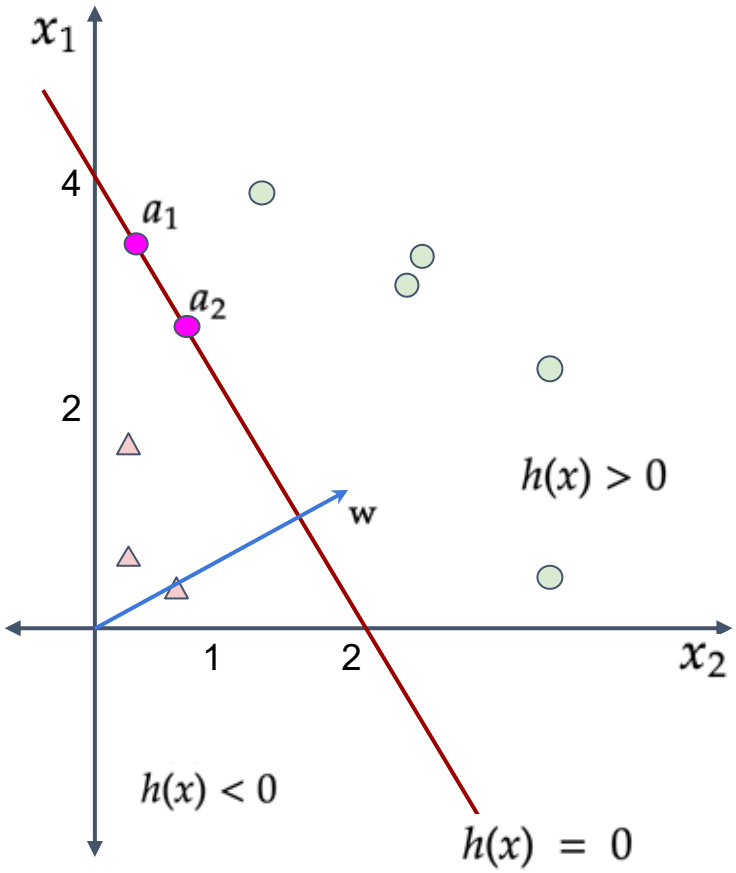
$$h(\mathbf{a}_1) = \mathbf{w}^T\mathbf{a}_1 + b = 0$$

$$h(\mathbf{a}_2) = \mathbf{w}^T\mathbf{a}_2 + b = 0$$

Subtracting one from the other:

$$\mathbf{w}^T(\mathbf{a}_1 - \mathbf{a}_2) = 0$$

The weight vector **w** is orthogonal to the hyperplane.



Distance of a Point to the Hyperplane

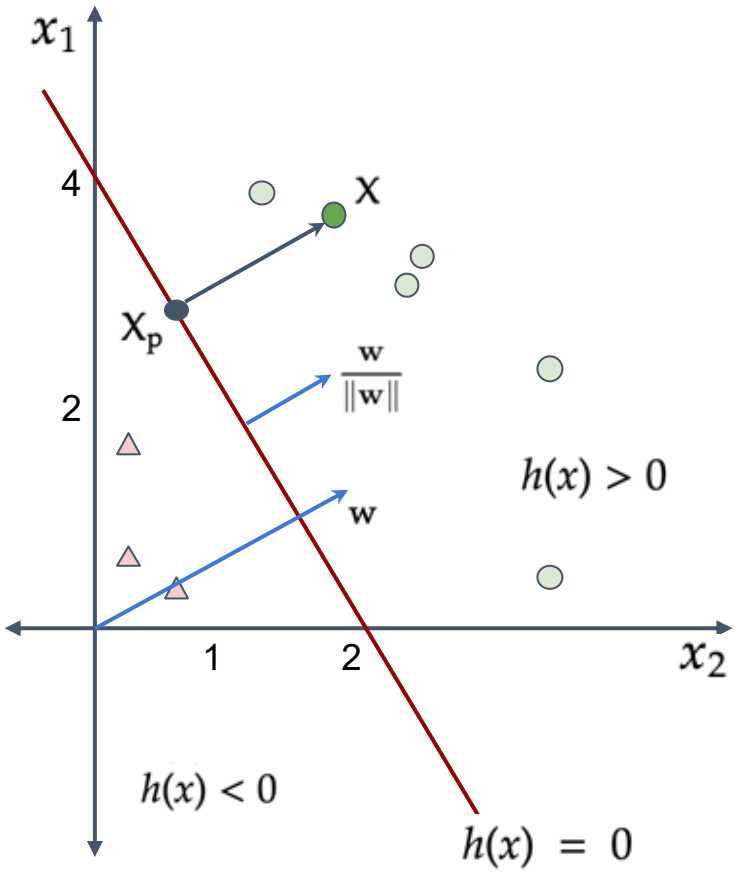
Consider a point X not on the hyperplane. Let X_p be the projection of X on the hyperplane.

Let r be the steps need to walk from X_p to X.

$$\mathbf{x} = \mathbf{x}_p + r\frac{\mathbf{w}}{\|\mathbf{w}\|}$$

$$\|w\| = \sqrt{w^T w} = \sqrt{\sum_i w_i^2}$$

Q: how many steps/direct distance do we need to walk?



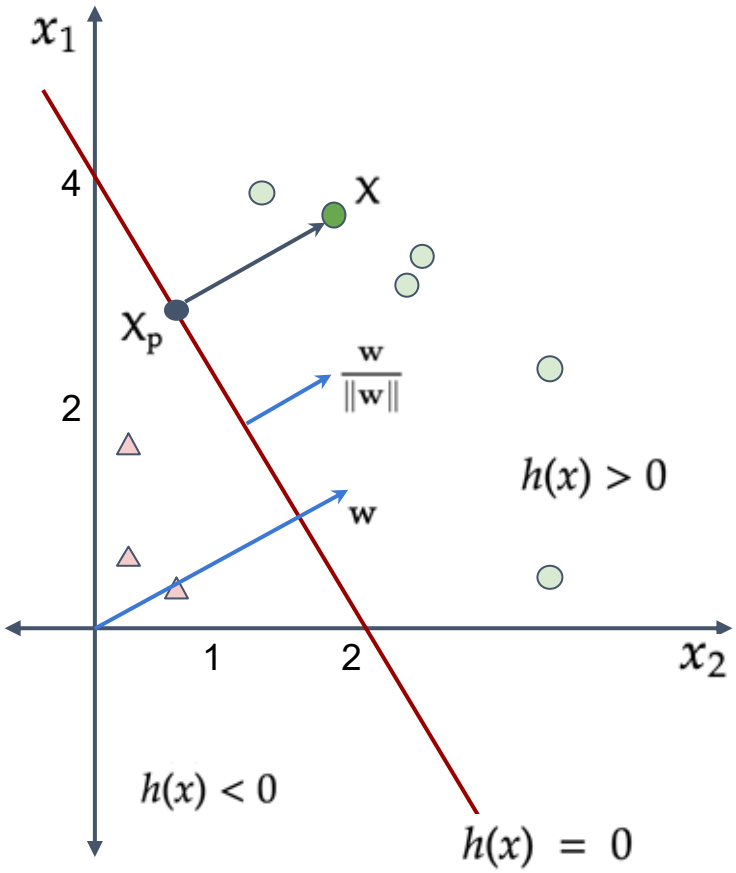
Distance of a Point to the Hyperplane

Consider a point X not on the hyperplane. Let X_p be the projection of X on the hyperplane.

Let r be the steps need to walk from X_p to X.

$$
\begin{aligned}
h(\mathbf{x}) &= h(\mathbf{x}_p + r\frac{\mathbf{w}}{\|\mathbf{w}\|}) \\
&= \mathbf{w}^T\left(\mathbf{x}_p + r\frac{\mathbf{w}}{\|\mathbf{w}\|}\right) + b \\
&= \underbrace{\mathbf{w}^T\mathbf{x}_p + b}_{h(\mathbf{x}_p)} + r\frac{\mathbf{w}^T\mathbf{w}}{\|\mathbf{w}\|} \\
&= \underbrace{h(\mathbf{x}_p)}_{0} + r\|\mathbf{w}\| \\
&= r\|\mathbf{w}\|
\end{aligned}
$$

$$r = \frac{h(\mathbf{x})}{\|\mathbf{w}\|}$$



Distance of a Point to the Hyperplane

Q: What is the direct distance from origin (x=0) to the hyperplane?

$$r = \frac{h(\mathbf{x})}{\|\mathbf{w}\|} \qquad r = \frac{h(\mathbf{0})}{\|\mathbf{w}\|} = \frac{\mathbf{w}^T\mathbf{0} + b}{\|\mathbf{w}\|} = \frac{b}{\|\mathbf{w}\|}$$

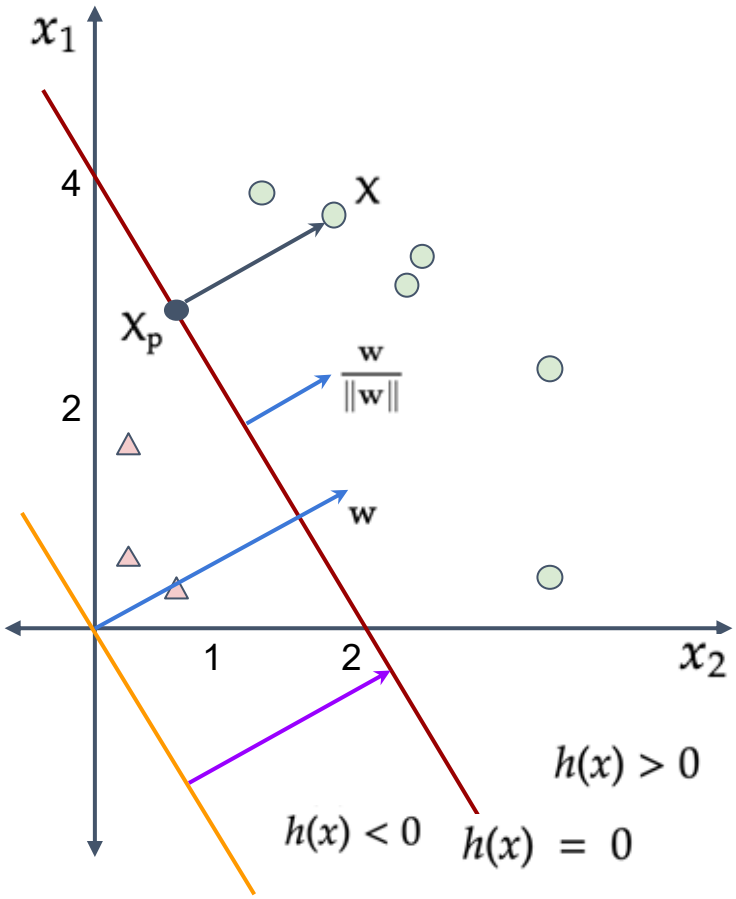
Example:

$$h(x) = x_1 + 2x_2 - 4$$

$$w^T x + b = (1 \;\; 2)\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} - 4$$

$$\frac{b}{\|w\|} = -\frac{4}{\sqrt{5}}$$

Q: how to deal with negative distance?



Distance of a Point to the Hyperplane

Q: How to deal with negative distance?

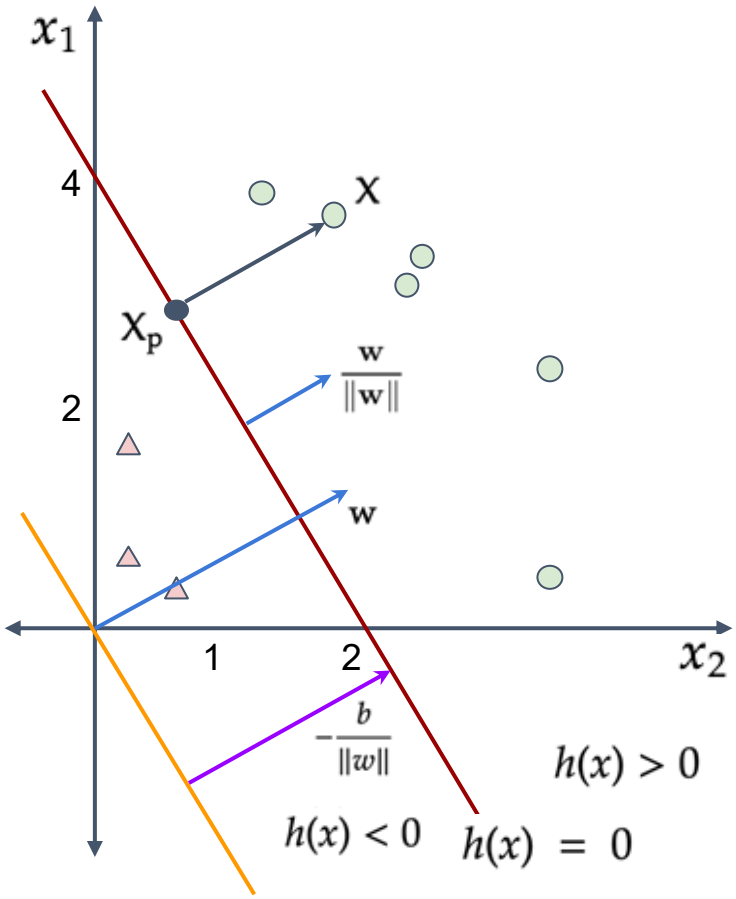
$$r = \frac{h(\mathbf{x})}{\|\mathbf{w}\|}$$

$$y = \begin{cases} +1 & \text{if } h(\mathbf{x}) > 0 \\ -1 & \text{if } h(\mathbf{x}) < 0 \end{cases}$$

$$\delta = y\, r = \frac{y\, h(\mathbf{x})}{\|\mathbf{w}\|}$$

Example (when point is the origin):

$$(-1) \cdot \frac{b}{\|w\|} = \frac{4}{\sqrt{5}}$$



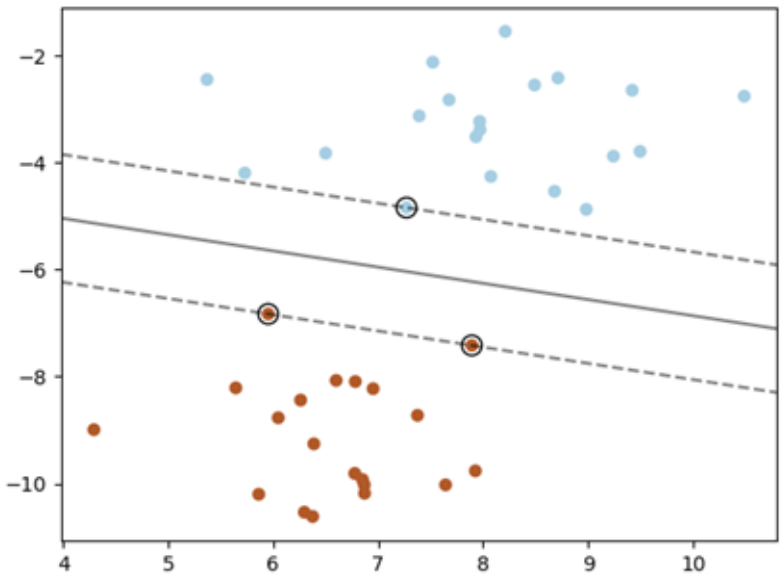
Margin and Support Vectors

$$\delta = y\, r = \frac{y\, h(\mathbf{x})}{\|\mathbf{w}\|}$$

Over all the n points, the ***margin*** of the linear classifier is the minimum distance of a point from the separating hyperplane:

$$\delta^* = \min_{\mathbf{x}_i} \left\{ \frac{y_i(\mathbf{w}^T \mathbf{x}_i + b)}{\|\mathbf{w}\|} \right\}$$

All the points that achieve this minimum distance are called ***support vectors***.



Max-Margin Classifier (Linear Separable Case)

For training data $\{(x^{(i)}, y^{(i)})\}_{i=1}^{m}$, a classifier $f(x) = w^\top x + b$ with 0 train error will satisfy

$$y^{(i)} f(x^{(i)}) = y^{(i)}(w^\top x^{(i)} + b) > 0$$

↓ negative margin when misclassifying it!

The distance for $(x^{(i)}, y^{(i)})$ to separating hyperplane

$$\frac{y^{(i)}(w^\top x^{(i)} + b)}{\|w\|}$$

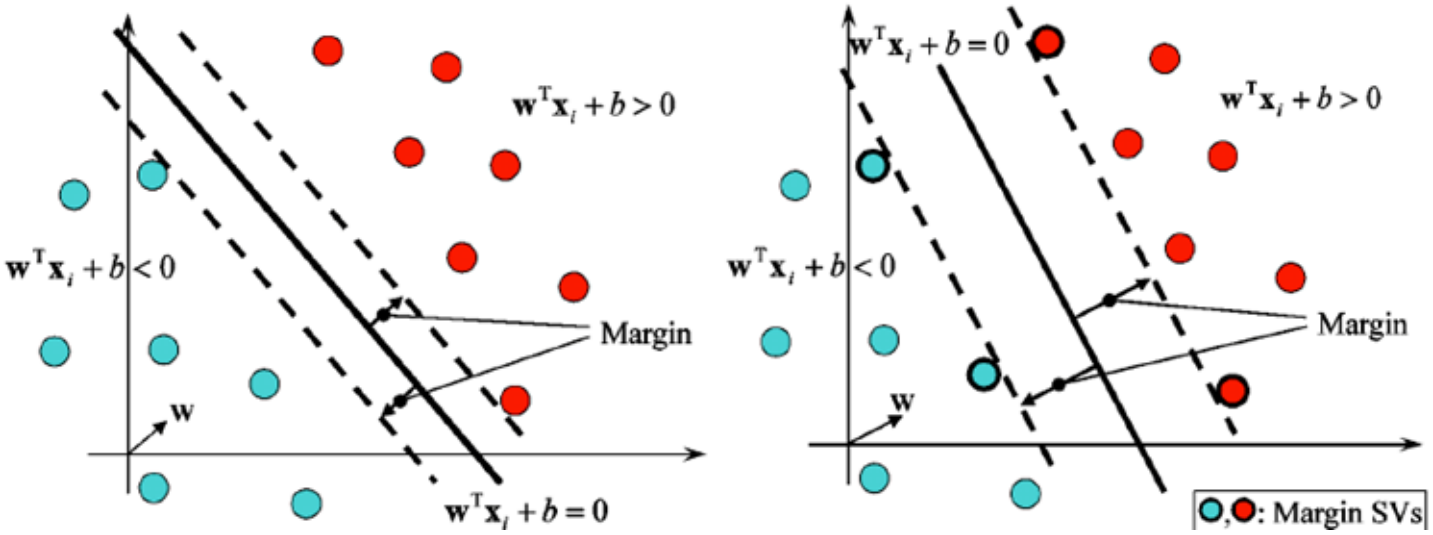
The margin of a classifier $f(x)$ is

$$\min_i \frac{y^{(i)}(w^\top x^{(i)} + b)}{\|w\|}$$

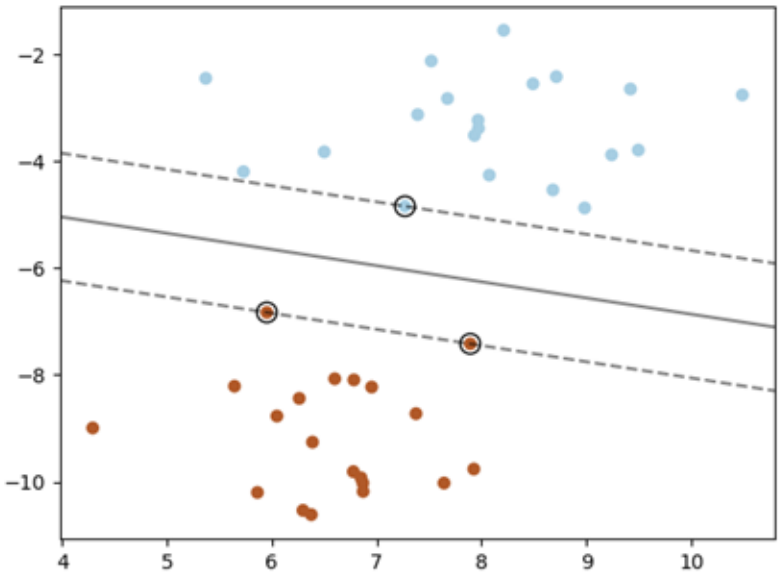
Find f that maximize margin

$$\arg\max_{w,b} \min_i \frac{y^{(i)}(w^\top x^{(i)} + b)}{\|w\|}$$

Max-Margin Classifier (Linear Separable Case)



Max-Margin Classifier (Linear Separable Case)



$$\overbrace{\arg\max_{w,b} \underbrace{\min_{i} \frac{y^{(i)}(w^{\top}x^{(i)} + b)}{\|w\|}}_{\text{Minimum margin over all training data}}}^{\text{Maximize the minimum margin}}$$

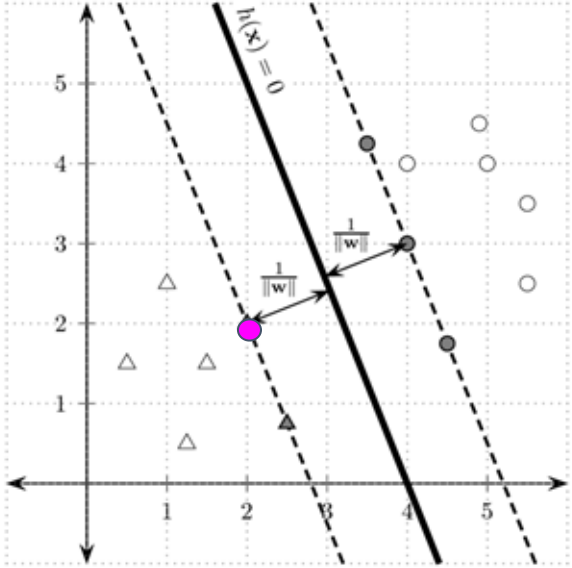
Find the parameters (w,b) that **maximize** the **smallest margin** over all the training data

Canonical Hyperplane

$$\arg\max_{w,b} \min_{i} \frac{\boxed{y^{(i)}(w^{\top}x^{(i)} + b)}}{\|w\|} \longrightarrow \text{1 for support vectors}$$

Issue: infinite equivalent hyperplanes result in infinite solutions

Solution: choose the scalar s such that the absolute distance of a ***support vector*** from the hyperplane is $\frac{1}{||w||}$

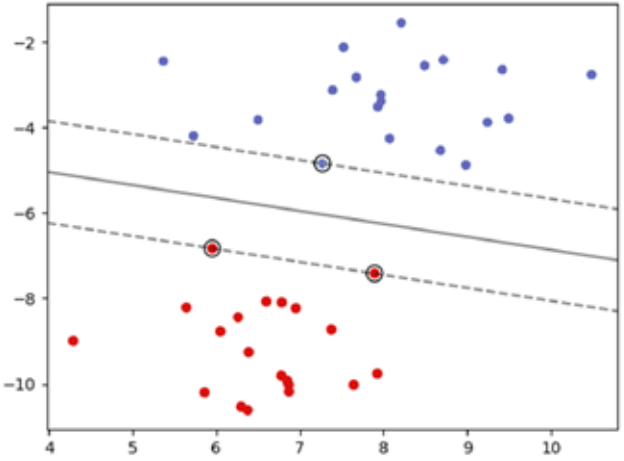


Margin: $\delta^* = \dfrac{1}{\|\mathbf{w}\|}$

Max margin: $h^* = \arg\max_{h} \{\delta_h^*\} = \arg\max_{\mathbf{w},b} \left\{ \dfrac{1}{\|\mathbf{w}\|} \right\}$

Support Vector Machine (Hard Margin)

… it leads to



$$\min_{w,b} \frac{1}{2}\|w\|^2$$

subject to

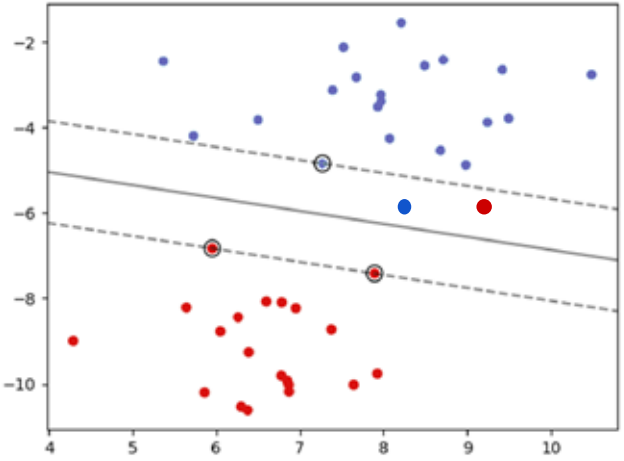
$$y^{(i)}(w^\top x^{(i)} + b) \geq 1 \qquad \text{for } i = 1, \ldots, m$$

This is a convex (quadratic) optimization problem that can be solved efficiently

- Data are D-dimensional *vectors*
- Margins determined by nearest data points called ***support vectors***
- We call this a *support vector machine* (SVM)

Support Vector Machine (Soft Margin)

If the data is linearly not separable,



regularization

error

$$\min_{w,b} \frac{1}{2}\|w\|^2 + C \cdot \sum_{i=1}^{m} \xi_i$$

subject to

$$y^{(i)}(w^\top x^{(i)} + b) \geq 1 - \xi_i$$

$$\xi_i \geq 0 \quad \text{for } i = 1, \ldots, m$$

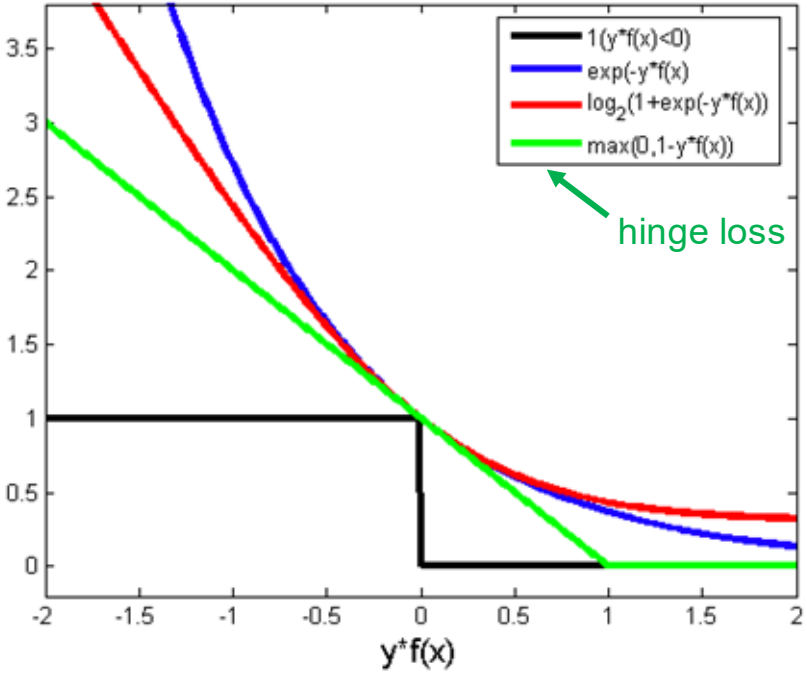
Equivalent formulation

$$\min_{\mathrm{w,b}} \frac{1}{2}\|w\|^2 + C \sum_{i=1}^{m} \left(1 - y^{(i)}(w^\top x^{(i)} + b)\right)_+$$

SVM - Soft Margin: hinge loss

Equivalent formulation

$$\min_{w,b} \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{m}\left(1 - y^{(i)}(w^\top x^{(i)} + b)\right)_+$$



$$\ell(f; x^{(i)}, y^{(i)}) = \left(1 - y^{(i)} f(x^{(i)})\right)_+$$

$$(X)_+ := \max(X, 0)$$

SVM - Soft Margin: an example

$$Hinge\ loss\ =\ max\left(0,\ 1-y_i\left(w^T x_i+b\right)\right)$$

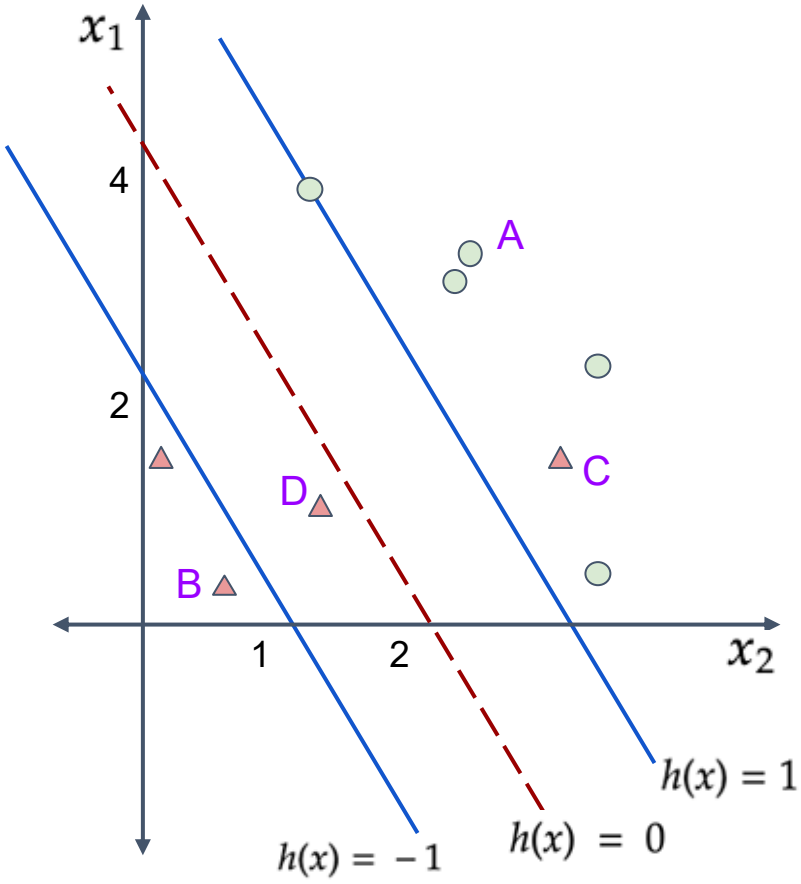
$$A:\ max(0,\ 1-1\cdot(>1))\rightarrow 0$$

$$B:\ max(0, 1-(-1)\cdot(<-1))\rightarrow 0$$

$$C:\ \ max(0,\ 1-(-1)\cdot(>1))\rightarrow\ >1$$

$$D:\ max(0, 1-(-1)\cdot(between\ [-1,0]))$$

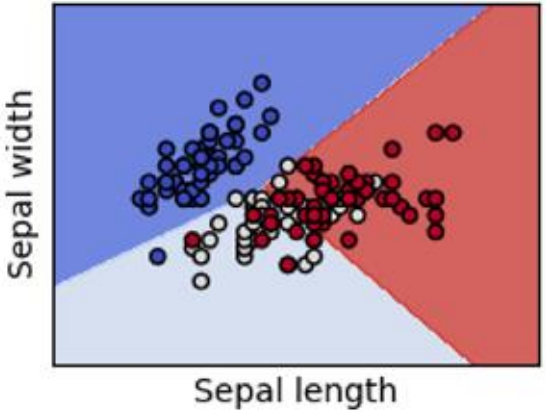
$$\rightarrow between\ [0,1]$$



SVM in Scikit-Learn

SVM with linear decision boundaries,

`sklearn.svm.LinearSVC`



Call options include…

penalty : *{'l1', 'l2'}, default='l2'*

Specifies the norm used in the penalization. The 'l2' penalty is the standard used in SVC. The 'l1' leads to `coef_` vectors that are sparse.

dual : *bool, default=True*

Select the algorithm to either solve the dual or primal optimization problem. Prefer dual=False when n_samples > n_features.

C : *float, default=1.0*

Regularization parameter. The strength of the regularization is inversely proportional to C. Must be strictly positive.

sklearn.svm.SVC

kernel : *{'linear', 'poly', 'rbf', 'sigmoid', 'precomputed'}, default='rbf'*

Specifies the kernel type to be used in the algorithm. It must be one of 'linear', 'poly', 'rbf', 'sigmoid', 'precomputed' or a callable. If none is given, 'rbf' will be used. If a callable is given it is used to pre-compute the kernel matrix from data matrices; that matrix should be an array of shape `(n_samples, n_samples)`.

gamma : *{'scale', 'auto'} or float, default='scale'*

Kernel coefficient for 'rbf', 'poly' and 'sigmoid'.

for RBF,
small γ: complex decision boundary
large γ: more like linear decision boundary

- if `gamma='scale'` (default) is passed then it uses 1 / (n_features * X.var()) as value of gamma,
- if 'auto', uses 1 / n_features.

max_iter : *int, default=-1*

Hard limit on iterations within solver, or -1 for no limit.

verbose : *bool, default=False*

Enable verbose output. Note that this setting takes advantage of a per-process runtime setting in libsvm that, if enabled, may not work properly in a multithreaded context.

class_weight : *dict or 'balanced', default=None*

Set the parameter C of class i to class_weight[i]*C for SVC. If not given, all classes are supposed to have weight one. The "balanced" mode uses the values of y to automatically adjust weights inversely proportional to class frequencies in the input data as `n_samples / (n_classes * np.bincount(y))`.

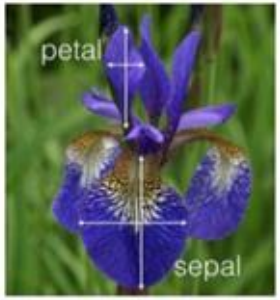
Example: Fisher's Iris Dataset

Classify among 3 species of Iris flowers…

Iris setosa

Iris versicolor

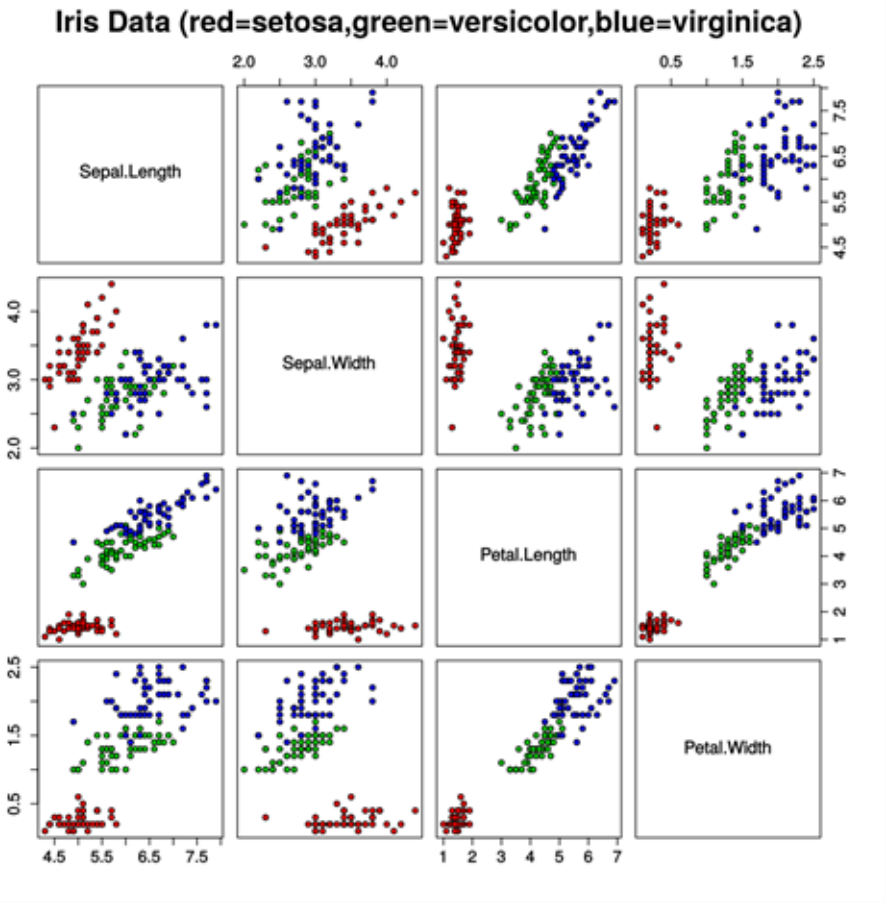
Iris virginica



Four features (in centimeters)

- Petal length / width
- Sepal length / width

Example: Fisher's Iris Dataset



Fairly easy to separate ***setosa*** *from others using a* *<u>linear classifier</u>*

Need to use nonlinear basis / kernel representation to better separate other classes

Example: Fisher's Iris Dataset

Train 8-degree polynomial kernel SVM classifier,

```
from sklearn.svm import SVC
svclassifier = SVC(kernel='poly', degree=8)
svclassifier.fit(X_train, y_train)
```

Generate predictions on held-out test data,

```
y_pred = svclassifier.predict(X_test)
```

Show confusion matrix and classification accuracy,

```
print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

```
[[11  0  0]
 [ 0 12  1]
 [ 0  0  6]]
```

	precision	recall	f1-score	support
Iris-setosa	1.00	1.00	1.00	11
Iris-versicolor	1.00	0.92	0.96	13
Iris-virginica	0.86	1.00	0.92	6
avg / total	0.97	0.97	0.97	30

[Source: https://stackabuse.com/implementing-svm-and-kernel-svm-with-pythons-scikit-learn/]